

FaceChain: Unifying AI Face Analysis, Reverse Image Search, and Blockchain Provenance

A Privacy-Aware Prototype for Verifiable Digital Media

Technical Project Report | Domain: Artificial Intelligence · Computer Vision · Blockchain
Built entirely with free and open-source tools | September 2026

Abstract

FaceChain is a privacy-aware prototype system that unifies three independently common technologies - AI-based face recognition, reverse image search, and blockchain-based provenance into a single verifiable pipeline. Given a user-provided image, the system detects and embeds any face present, searches an authorized image collection for visually or facially similar entries using vector similarity search, and generates a cryptographically verifiable provenance record for every registered image using a hash-linked local blockchain. The result is a lightweight tool for authorized digital media verification: it can confirm whether an image already exists in a trusted collection, how similar it is to existing entries, and whether that image's content has been altered since it was first registered. The entire system was implemented using free and open-source tools, making it suitable as an academic prototype or a foundation for further development toward a production-grade, testnet-deployed provenance service.

Keywords: *Face Recognition, Reverse Image Search, Vector Similarity, FAISS, Blockchain Provenance, SHA-256, Digital Media Verification*

1. Introduction

Digital images are trivial to copy, alter, and redistribute, which creates ongoing challenges for media verification, academic dataset integrity, and intellectual property protection. Existing solutions typically address only one part of this problem: conventional face recognition systems compare faces but do not verify image integrity; reverse image search tools locate duplicates but ignore facial similarity; and provenance tools (such as C2PA-style hashing) verify integrity but perform no content-based search.

FaceChain combines all three approaches into a single working system, allowing a user to answer three questions from one upload: does this face resemble a known face, does this image resemble a known image, and can the image's history be verified.

2. Tools and Technologies Used

All components of FaceChain were built using free, open-source software, requiring no paid APIs or cloud services during development or testing. Table 1 summarizes the technology stack across each layer of the system.

Table 1: Tools and Technologies Used

Category	Tools / Technologies (all free / open-source)
Programming language	Python 3.11 (backend, AI logic), JavaScript / HTML / CSS (frontend)
Face detection & embedding	DeepFace (Facenet model), OpenCV, TensorFlow
Similarity search	FAISS (Facebook AI Similarity Search) - vector indexing and cosine similarity
Backend framework	FastAPI with Uvicorn ASGI server
Blockchain / provenance	Python hashlib (SHA-256), custom hash-linked local blockchain (JSON-based ledger)
Frontend	HTML5, CSS3, vanilla JavaScript (Fetch API), responsive dashboard UI
Development tools	Visual Studio Code, Git, Python venv, pip

3. System Architecture and Working Flow

The system follows a linear, eight-stage pipeline from image upload to verification, illustrated in Fig. 1. Each stage is handled by a dedicated backend module, keeping face analysis, similarity search, and blockchain logic independent and testable.

3.1 Module Breakdown

- **face_module.py:** wraps DeepFace to detect a face and produce a 128-dimensional embedding, trying multiple detector backends for robustness.
- **search_module.py:** maintains a FAISS vector index of all registered embeddings and returns ranked similarity results.
- **blockchain_module.py:** generates SHA-256 fingerprints and maintains a hash-linked chain of provenance blocks, each referencing the previous block's hash.
- **main.py:** a FastAPI application exposing /register, /search, /verify, and /status endpoints that tie the above modules together.
- **frontend (index.html):** a browser-based dashboard for uploading images and viewing results, communicating with the backend via the Fetch API.

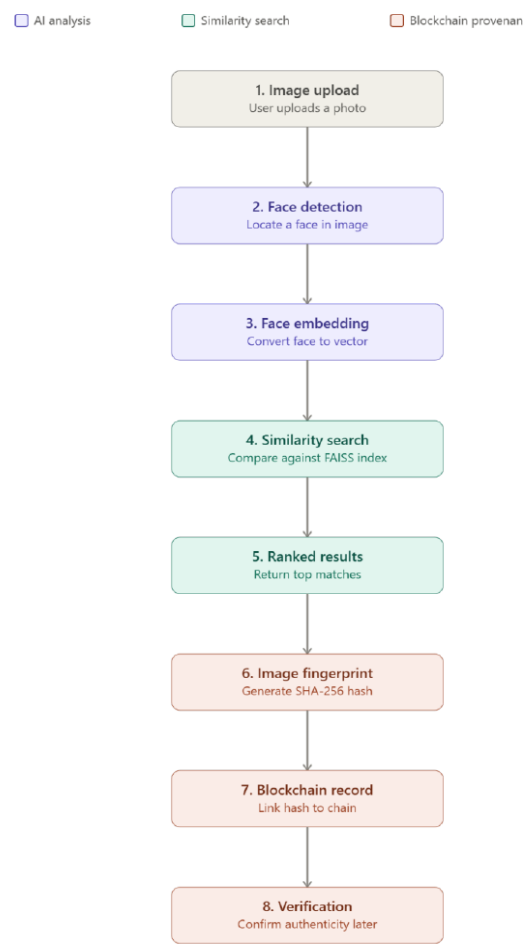


Fig. 1: Architecture and Workflow of the FaceChain Detection System

Table 2: Eight-Stage Pipeline

Step	Stage	Description
1	Image Upload	User uploads a JPG/PNG image through the dashboard.
2	Face Detection	DeepFace detects the presence and location of a face in the image.
3	Face Embedding	The detected face is converted into a 128-dimension numerical vector (Facenet).
4	Similarity Search	FAISS compares the embedding against all authorized images using cosine similarity.

5. Discussion

The results confirm that combining face embeddings with vector similarity search produces intuitive, well-separated similarity scores: identical images score at the maximum (1.00), visually or facially related images score in a moderate range, and unrelated images score near zero.

The blockchain layer, while implemented as a local hash-linked ledger rather than a public network, demonstrates the same tamper-evidence property as a public blockchain: any modification to a stored record would break the hash chain and be immediately detectable during validation. This design keeps the prototype free to build and run while preserving the conceptual integrity of the original blockchain-provenance requirement.

6. Conclusion and Future Scope

FaceChain successfully demonstrates that AI-based face analysis, reverse image search, and blockchain provenance can be unified into a single, coherent, and free-to-build system. The prototype fulfils its intended use cases - digital media verification, academic dataset tracking, and authenticity checking within a controlled, authorized image collection.

Planned Future Improvements

- Deploying the provenance layer to a public Ethereum test network (e.g. Sepolia) via a free RPC provider, replacing the local JSON ledger with genuine decentralized storage.
- Adding user authentication so only authorized accounts can register new images.
- Expanding the authorized collection and benchmarking search performance at larger scale.
- Extending reverse image search beyond faces to general visual similarity for non-facial images.

In summary, FaceChain uses AI to compare, reverse image search to find, and blockchain to verify the history and integrity of authorized digital images achieved entirely with free and open-source tools.